Department of Computer Science

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Introduction of the Project

The Smart Parking System is a digital solution designed to automate the management of vehicle parking areas. Developed in C, this application aims to replace manual entry logs with an efficient, real-time computerized system that tracks vehicle status and space availability.

Project Objective

The primary objective is to develop a reliable system that can:

- Monitor the occupancy of parking slots in real-time.
- Reduce the time spent by drivers looking for vacant spots.
- Minimize manual errors in record-keeping.

Background Problem

Traditional parking management often relies on manual labor, leading to traffic congestion at entry points and difficulty in tracking available spaces. Furthermore, manual data entry is prone to human error and makes retrieving historical parking records inefficient.

Project Purpose

The purpose of this project is to provide a structured interface where:

- Admins can easily check the total number of parked vehicles.
- The system automatically generates entry and exits timestamps.
- Automated billing is calculated based on the duration of the stay.

Languages & Technologies Used

 Language**:** C Programming Language.

 Concepts**:** Arrays/Structures, File Handling (for data persistence), and Conditional Logic.

 Environment**:** Dev C++

Project Features

 Dynamic **Slot Allocation:** Assigns specific slots to vehicles.

 Real**-time Status:** Displays current occupancy levels.

 Automated **Billing:** Calculates fees upon vehicle exit.

 Search **Function:** Allows admins to find specific vehicle records by plate number.

Plan Of Work

 Phase **1:** Problem Analysis and System Design.

 Phase **2:** Developing the console-based UI and Core Logic.

 Phase **3:** Implementing File Handling to save parking logs.

 Phase **4:** Testing the system for bugs and finalizing documentation.

Conclusion

The Smart Parking System simplifies the complexities of modern parking management. By leveraging C language's speed and file handling capabilities, the system ensures a fast and reliable experience for both administrators and users.

Future Expansion

In the future, this system could be integrated with IoT sensors to detect physical vehicle presence or updated with a Graphical User Interface (GUI) for a more interactive experience.